

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

1 . Power (P) in the headlight bulb can be calculated using the formula $P = IV$, where P is power in watts , I is current in amperes , and V is voltage in volts . The power is given as 40 watts and the voltage of the battery is 12 volts .

2 . Rearrange the formula to solve for current : $I = P / V$. Using the given power and voltage , we can calculate the current .

3 . Once we have the current , we can use the power formula ($P = IV$) again to find the resistance (R), but we already know P and I , so $P = IV = R * I$, which can be rearranged to $R = P / I$.

4 . Substitute the known values into these equations to get the current and resistance .

</think>

<answer>

B . 2 . 8 amps , 3 . 2 ohms

</answer>